

CONTACT & PROFILES

- **Email:** ashikur.gub@gmail.com
- **Phone:** +880 1962-579305
- **Portfolio:** favouritesir.me
- **GitHub:** github.com/favouritesir
- **LinkedIn:** linkedin.com/in/favouritesir
- **Location:** Gazipur / Dhaka, Bangladesh

CORE TECHNICAL SKILLS

Algorithms & Low-Level

C / C++ | Pointer In-Situ | Asymmetric Space | ATmega328P

Languages & Runtimes

JavaScript (ES6+) | TypeScript | Node.js | Python | Java | Bash / Shell

Web & Frameworks

React.js | Vue.js | Electron.js | Three.js | HTML5 / CSS3

Cloud, Serverless & Data

Google Apps Script | Real-Time Sync | MySQL | Prisma ORM | Git / GitHub

LANGUAGES

Bengali:
Native Proficiency

English:
Full Professional Working

ASHIKUR RAHMAN

SOFTWARE ENGINEER • ALGORITHM RESEARCHER • SYSTEMS ARCHITECT

Computer Science & Engineering graduate from Green University of Bangladesh. Conceived, proved, and published novel algorithmic paradigms indexed with permanent CERN/Zenodo DOIs. Architect of zero-cost, high-concurrency real-time cloud management platforms, developer NPM utilities, and embedded IoT networking firmware. Proven track record in academic mentorship, systems operations, and production software delivery.

PUBLISHED RESEARCH PREPRINTS (CERN / ZENODO DOIS)

Cloud Sort: An Overhead Caller-Buffer Merge Paradigm

DOI: 10.5281/zenodo.23665548

Conceived at Dept. of CSE, Green University of Bangladesh • Published Oct 2026

Invented an asymmetric in-situ merge sort reducing auxiliary space to [N/2] elements with zero dynamic heap allocations (`malloc/free`). Proved inductive read-before-write safety ($wp_t \le rp_t$) and formulated the **Zero-Move Right Tail Property**, eliminating 25.0% of physical SRAM data bus writes. Audited against QuickSort, Standard Merge, and TimSort: achieves 10%–67% faster throughput than textbook merge with 0% extra key comparisons. Verified across 10,000 differential random array trials and micro-benchmarked on ATmega328P.

In-Situ Dual-Pointer Algorithm for Infix-to-Prefix Transformation

DOI: 10.5281/zenodo.23657544

Conceived at Dept. of CSE, Green University of Bangladesh • Published Oct 2026

Formulated an optimal right-to-left single-pass conversion algorithm achieving $O(1)$ auxiliary string space. Eliminates conventional textbook triple-pass string reversal overhead, reducing memory bus operations by over 60% with full mathematical proofs.

FEATURED PRODUCTION & SYSTEMS PROJECTS

Demo PMS & AdisFee Cloud Platform (Live Serverless WebApp)

demopms.blogspot.com | adisfee.blospot.com

- Architected a high-concurrency student accounting and fee collection platform powered by Google Apps Script serverless backends with zero recurring cloud hosting costs.
- Engineered instant real-time state synchronization across independent browser tabs without WebSockets or persistent socket servers.
- Integrated cryptographic OTP-based authentication and role-based operational permissions.

SUPER Fill — Form Engine & Dynamic Workspace (IDP 1 Lab Project)

Course: CSE 324, Green University of Bangladesh

- Designed a cloud-native platform integrating a robust form engine with a dynamic Notion-style collaborative workspace for interactive quizzes, automated grading, and tasks.
- Authored complete architectural specifications: DFD Level 0/1, UML sequence & class diagrams, high-fidelity UI vector mockups, and comprehensive engineering reports.

Json-Boundary — NPM Package (TypeScript / Node.js)

npmjs.com/package/json-boundary

- Created an ultra-fast zero-dependency TypeScript library for scanning boundary indexes in JSON payloads, conceived to automate dynamic Prisma ORM `where` clause generation.

CERTIFICATIONS & HONORS

Itransition Group (Nov 2024)
Commercial Software Development — JavaScript (Corporate Remote Training)

CERN / Zenodo Verified Author
2x Published DOIs for Original Data Structures & Algorithms Research

Founder • BD Smart Book
Online Publication & Automated Learning System

EMBEDDED & SYSTEMS

ESP8266 / ESP32 | DNS Protocols | Linux POSIX | Terminal ANSI TUI | Process Control

ENGINEERING PHILOSOPHY

"True software excellence balances theoretical rigor with physical execution — eliminating memory waste, zeroing unnecessary allocations, and building resilient zero-cost systems."

ONLINE REPOSITORY

Full source codes, benchmarks, test vectors, and multi-language ports available on GitHub:
github.com/favouritesir

SYSTEMS, IOT & AUTOMATION PROJECTS (CONTINUED)

My-mini-DNS — Custom ESP Microcontroller DNS Router

`github.com/favouritesir/My-mini-DNS`

- Engineered custom local DNS firmware in C++ for ESP8266/ESP32 microcontrollers.
- Enables dual-routing capability: intercepts and resolves private local domain names to local micro-services while transparently passing upstream internet queries to public DNS servers.

mTask — Interactive Terminal Task Manager (Bash / Linux)

`github.com/favouritesir/mTask`

- Developed a terminal-based responsive task dashboard in pure POSIX Bash for Linux OS coursework.
- Implemented keyboard-driven navigation, ANSI escape graphics, task prioritization, and background process tracking with zero external dependencies.

Electron.js Task Manager • Don Entry Automation

Cross-Platform Desktop & Cloud Ledger

- **Electron Task Manager:** Built cross-platform desktop task suite with persistent local storage.
- **Don Entry:** Automated financial donation and ledger entry pipelines on Google Apps Script with public multi-part video documentation (YouTube Playlist).

PROFESSIONAL EXPERIENCE & LEADERSHIP

Technical Advisor & Academic Systems Lead (7+ Years)

2017 — Present

Alor Dishary Ideal School • Gazipur, Bangladesh

- Architected and managed computerized institutional grading systems, digital student records, and automated administrative fee collection systems.
- Designed specialized academic curricula, interactive question sheets, and STEM learning materials.
- Taught advanced Computer Science, Physics, Chemistry, and General Mathematics, mentoring hundreds of students through secondary academic cycles.

Founder & Platform Architect

2021 — Present

BD Smart Book (bdsmartbook.com)

- Founded an online educational publishing platform providing digitized smart learning materials.
- Engineered interactive smart books automated via Google Apps Script and modern responsive web interfaces.

Commercial Software Development Trainee

Nov 2024

Itransition Group (Remote Corporate Training)

- Completed intensive corporate training in commercial JavaScript/TypeScript development, mastering enterprise-level coding practices, code reviews, and production software patterns.

EDUCATION

Degree / Certificate	Institution	Group / Department	Year / Result
B.Sc in CSE (Completed)	Green University of Bangladesh	Computer Science & Engineering	4-Year (8 Semesters)
H.S.C (Higher Secondary)	Gazipur City College	Science	2018 • GPA: 3.33 (Dhaka)
S.S.C (Secondary School)	Gazipur City Model School	Science	2016 • GPA: 4.78 (Dhaka)